VoiceKind

Voice & Gesture Controlled Interaction System

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THE MISSION

Designed for Inclusivity

VoiceKind is an intelligent accessibility system that enables seamless **hands-free control** of computers and browsers. By leveraging AI-powered recognition, we empower users to navigate the digital world through natural human movement and speech.

SPEECH INTERACTION

- "Scroll Down / Up"
- "Click on Login"
- "Search for Headphones"
- "Reload Page / Go Back"

Dual Mode Logic

Powered by Google SpeechRecognition API for high accuracy, with an offline fallback mode using **Vosk** for privacy and reliability.

VISION-BASED GESTURES

Cursor Tracking
Index finger positioning tracked in real-time for fluid navigation.

Trigger Click
Closing the fist triggers a mouse click event instantly.

Smooth Scroll
Vertical hand movements mapped to page scrolling speeds.

HYBRID ARCHITECTURE

Component	Role	Tech Used
Python Backend	Core logic for voice + gesture recognition	OpenCV, Mediapipe
Chrome Extension	Direct DOM & Web interaction	JavaScript (Manifest V3)
WebSocket Server	Real-time low-latency data bridge	Python WebSocket

THE ENGINEERING STACK

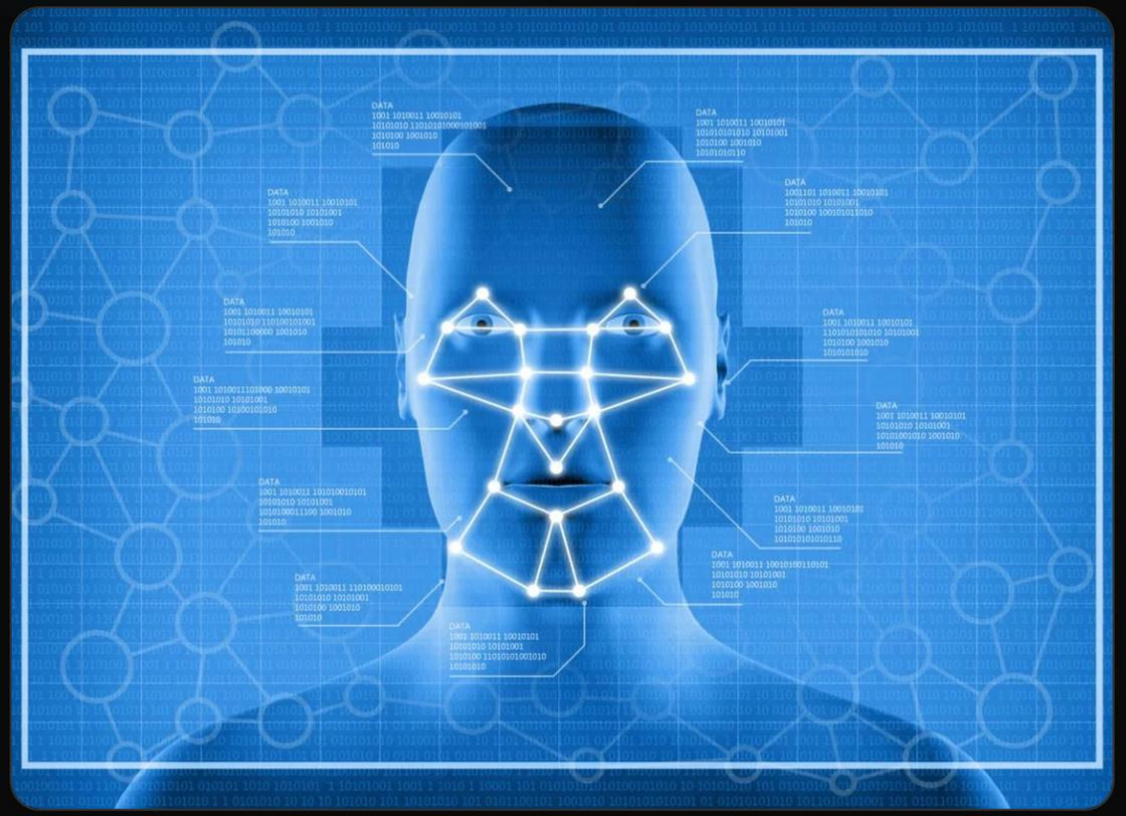
Backend (Python)

- OpenCV: Frame Processing
- Mediapipe: Hand Landmark Tracking
- PyAutoGUI: System-level Automation

Frontend (Web)

- JavaScript: Browser Manipulation
- Manifest V3: Secure Extension Framework
- DOM Interaction: Elements control

MULTIMODAL INTEGRATION



Parallel Processing

VoiceKind is unique because it allows **gesture and voice to work simultaneously**. Users can move the cursor with their hands while issuing voice commands to click or search, creating a truly natural interactive loop.

REAL-WORLD IMPACT

Accessibility
Empowering those with motor impairments to use PCs independently.

Sanitary Control
Hands-free usage in medical or kitchen environments.

Presentations
Controlling large displays from a distance without a mouse.

DEVELOPMENT ROADMAP

1. Integration of Deep Learning for custom gesture training.
2. Multi-language support for voice commands.
3. Low-power optimization for mobile browser support.

Thank You

Questions? Let's make the web accessible for everyone.

Raveendra Gunapu
Project: VoiceKind

IMAGE SOURCES

https://socialnomics.net/wp-content/uploads/2025/08/facial-recognition-system.png

Source: socialnomics.net